



Reflections of Life: Distinguishing Living from Nonliving Worlds with Complexity Metrics

Stuart Bartlett^{1,2} , Xianlei San^{1,3} , Siteng Fan³ , Gary Segal⁴ , Damian R. Sowiński⁵ , and Yuk L. Yung^{1,6}

¹ Division of Geological & Planetary Sciences, California Institute of Technology, Pasadena, CA, USA

² SETI Institute, Mountain View, CA, USA

³ Southern University of Science & Technology, Shenzhen, People's Republic of China; fanst@sustech.edu.cn

⁴ University of Queensland, Queensland, Australia

⁵ Department of Physics & Astronomy, University of Rochester, Rochester, NY, USA

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Abstract

Detecting life beyond Earth is the primary goal of astrobiology. In this study, we showcase the effectiveness of information and complexity-based metrics for detecting signs of life on exoplanets. These metrics are applied to time series of disk-integrated light reflectance from Earth and its close relative, Mars, a planet predicted to have once been very similar to Earth. The light curves used in this study are analogous to those obtainable for exoplanets situated at great distances, hence we are treating Earth and Mars as proxy exoplanets. We found that Earth scores higher across all information-theoretic and complexity-based metrics employed, compared to Mars. These include the Shannon entropy, zip compressibility, joint differential entropy of eigencolours, statistical complexity and epsilon machine entropy rate. This supports the hypothesis that planetary complexity, as measured from reflectance time series, could be a promising indicator of life. The method does not rely upon chemical properties or in situ analyses, and could be applied to data from current and upcoming missions such as the James Webb Space Telescope and Habitable Worlds Observatory.

Unified Astronomy Thesaurus concepts: [Astrobiology \(74\)](#); [Biosignatures \(2018\)](#); [Light curves \(918\)](#); [Exoplanet astronomy \(486\)](#); [Exoplanet atmospheres \(487\)](#)

1. Introduction

Detecting life beyond Earth is a primary science goal for NASA, the field of astrobiology, and fundamental for understanding humanity's place in the universe (National Academies of Sciences, Engineering, and Medicine 2023a, 2023b). Our ability to reach beyond Earth has expanded greatly in recent years, and upcoming missions will be able to directly image exoplanets and monitor them for extended periods.

Searching for life is now a sub-discipline in its own right (J. L. Grenfell 2017; S. I. Walker et al. 2018; B. Cavalazzi et al. 2019; N. Guttenberg et al. 2021; H. J. Cleaves et al. 2023), and the majority of biosignature programs focus upon chemical properties, including so-called “biosignature gases” (V. S. Meadows et al. 2018; E. W. Schwieterman et al. 2018), the detection of biomolecules (M. A. Chan et al. 2019; D. J. Des Marais & L. L. Jahnke 2019; M. Dannenmann et al. 2023), chiral excess (D. P. Glavin et al. 2019; M. Gleiser 2022), or molecular complexity (S. M. Marshall et al. 2021). In parallel, there is an additional suite of techniques under the guise of “agnostic biosignatures,” which aim to be minimally biased toward life as we know it (S. S. Johnson et al. 2018; L. Chou et al. 2021; N. Guttenberg et al. 2021; D. Kinney & C. Kempes 2022; H. J. Cleaves et al. 2023; S. Vannah et al. 2024). An approach was recently proposed that aims to take this philosophy one step further (S. Bartlett et al. 2022), by avoiding chemical features entirely and instead focusing on the

hypothesis that life could be fundamentally associated with complexity, broadly defined (B. H. Weber 2010; J. P. Crutchfield 2012; H. Zenil et al. 2012; K. D. Farnsworth et al. 2013; C. H. Lineweaver et al. 2013; G. Tkačik & W. Bialek 2016; L. F. Seoane & R. V. Solé 2018). Here we suggest that such an approach is sufficiently powerful to distinguish the one inhabited planet in the solar system, Earth, from its close relative that was once very similar, Mars.

Complexity has been notoriously difficult to quantify (C. H. Bennett 1990; J. Horgan 1995), but great progress has been made with specific metrics including the Kolmogorov or algorithmic complexity (A. N. Kolmogorov 1963; H. Zenil et al. 2023), logical depth (C. H. Bennett 1988), and the field of computational mechanics J. P. Crutchfield (2012) and C. R. Shalizi & J. P. Crutchfield (2001). The common thread among these approaches is that complexity is related to the information content of the minimal, optimal representation of a given process or set of observations. While it is not possible to calculate the exact complexity of an object or process (due to certain mathematical constraints, such as the Halting Problem A. Church 1936; H. Zenil et al. 2023), algorithms do exist for computing the size of a “best” (subject to certain constraints) model of the observations of a system (J. P. Crutchfield 2012).

Signals from planets exhibit variability over a range of timescales. This can be quantified using the complexities of the time series of reflected light from these planets (the light from the host star reflected by the planet, resolved over a finite period of time, at a given sampling rate). When viewed from a great distance, planets can only be observed as single-point variations of light (no spatial detail or resolution), but such measurements can be made at multiple wavelengths or wavelength bands. Data of this form from the Deep Space Climate Observatory (DSCOVR) has been used to evaluate

⁶ Deceased.



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signs of life on Earth, treating our home as a proxy exoplanet (S. Fan et al. 2019; L. Gu et al. 2021, 2022; S. Bartlett et al. 2022). That data was also reconfigured into a variety of artificial permutations of reduced surface types (removing clouds, replacing the entire surface pixels with only desert pixels, etc.). It was found that both the Shannon entropy and “statistical complexity” (J. P. Crutchfield 2012) were able to cluster the synthetic time series into groups of associated planetary surface types. Because Earth has such depth of variation in its surface types and is highly dynamic, its light curves exhibited significantly higher complexities than better mixed gas giants such as Jupiter. More recent work showed that removing redundancies between wavelengths reveals an additional layer of distinctions between planetary surface types (G. Segal et al. 2024).

In the present work, we further test the hypothesis that informational and complexity metrics applied to reflectance time series can distinguish living from nonliving planets. We compared reflectance data from Earth and Mars, the two planets of highest astrobiological significance in the solar system. The data for both planets are from deep space observations: at the first Sun–Earth Lagrangian point of Earth for DSCOVR, and in a high-altitude orbit obtained by the Emirates Mars Mission (EMM) for Mars (H. Almatroushi et al. 2021; Emirates Mars Mission 2024). Our hypothesis that Earth’s reflectance complexity is higher than sterile worlds is supported by these new results, demonstrating the power of exoplanet reflectance time series for distinguishing biotic from abiotic worlds. The approach does not rely on chemical or molecular assumptions about life, functions without in situ measurements or samples, and can be applied to future missions such as the Habitable Worlds Observatory (HWO).

Our input data consisted of light reflectance time series for Earth and Mars (see Section 2.1). Among the various wavelengths available within this data, three wavelength values were sufficiently close across both planets to allow direct comparison. The three sets were 325 and 320 nm, 443 and 437 nm, and 551 and 546 nm, for Earth and Mars, respectively. These time series were processed so that they had equivalent sampling rates and lengths, both in terms of the number of data points and the duration, in units of local days. They were then individually normalized to the unit interval, and discretized (see Section 2.2). The analysis consisted of computing the Shannon entropy (C. E. Shannon 1948), Zip compressibility ratio (H. Zenil et al. 2023), “joint differential entropy of eigencolours” (JDE; G. Segal et al. 2024), and the statistical complexity and entropy rates of epsilon machines (J. P. Crutchfield 2012; see Section 2.3). The overall pipeline for our analysis is shown in Figure 1.

In the following section we detail the methodological details of our approach, present results in Section 3, and draw conclusions in Section 4.

2. Methods

2.1. Generation of Reflectance Time Series

Earth reflectance time series were sourced from the DSCOVR mission (J. Burt & B. Smith 2012). Full details of how these time series were constructed can be found in our previous publications (J. H. Jiang et al. 2018; L. Gu et al. 2021; S. Bartlett et al. 2022).

Mars time series were sourced from the EMM spacecraft, which entered Martian orbit on 2021 February 9, offering a global view of Mars from a near-equatorial, highly super-synchronous orbit with an inclination of 25 deg (H. Almatroushi et al. 2021). The periapsis of this orbit is approximately 20,000 km, apoapsis about 43,000 km, and the orbital period is 55 hr (H. Amiri et al. 2022). The Emirates Exploration Imager (EXI) is one of three instruments on EMM (Emirates Mars Mission 2024). EXI is a medium-resolution imaging system capable of capturing 12 megapixel images of the entire Martian disk using three ultraviolet (centered at 220, 260, 320 nm) and three visible light bands (centered at 437, 546, 635 nm), providing a spatial resolution of 2–4 km per pixel (A. M. Fernando et al. 2024).

EXI data are available at different levels: L1, L2A, L2B, L3, and Quicklooks, with our data selected from L2A. Level 2A files are first calibrated, then signal bias is estimated and subtracted, then the files are corrected for flat field effects. EXI data also provide data quality flags to remove poor quality or defective pixels. To obtain the full-disk intensity data of Mars, the height of the tangent point above the reference ellipsoid is determined to be zero (EXI Data Product Guide).

We integrate the spatially resolved images over the Martian disk to simulate observations of an exoplanet detected as a point source. At each time step, reflectance images are obtained sequentially from the five channels, with exposure times ranging from 22 to 654 ms. These are combined to form a five-point reflection spectrum. To maintain consistency and richness in the temporal span of the data during complexity analysis, we selected the data between the two dotted red lines in Figure 2.

Using the formula below, we corrected for distance, converting the original data into reflectance, shown in Figure 3:

$$A_L = \frac{\pi \cdot \text{radiance}}{\cos(i) \cdot \text{SOLFLUX}},$$

where radiance is calculated as described; SOLFLUX is the incident solar irradiance value found in the header; and i is the solar incident angle taken from the L2A HDU INA.

As shown in Figure 2, we observed that when the phase angle (PHA) exceeds 90°, the reflectance shows minimal variation, deviating from the predictions of Lambert’s reflectance theory. Therefore, we limited the PHA to values below 90° for subsequent data analysis. Using a third-order polynomial fit, we derived a function relating reflectance to the cosine of PHA (Figure 4). All data were subsequently converted to the full phase case (Figure 5), with the uncertainty σ_{total} calculated as:

$$\sigma_{\text{total}} = \sqrt{\sigma_{\text{meas}}^2 + \sigma_{\text{resp}}^2 + \sigma_{\text{HST}}^2 + \sigma_{\text{bin}}^2}. \quad (1)$$

Here, the measurement uncertainty (σ_{meas}) was derived from the standard deviation of reflectance values across all pixels within each observation. The response uncertainty (σ_{resp}), the uncertainty in the HST flux standards (σ_{HST}), and the binning factor (σ_{bin}) are reported in the EXI In-Flight Radiometric Calibration Technical Memo.⁷

⁷ https://s3.eu-west-1.amazonaws.com/emm.sdc-documentation/exi/EXI_In-Flight_Radiometric_Calibration.pdf

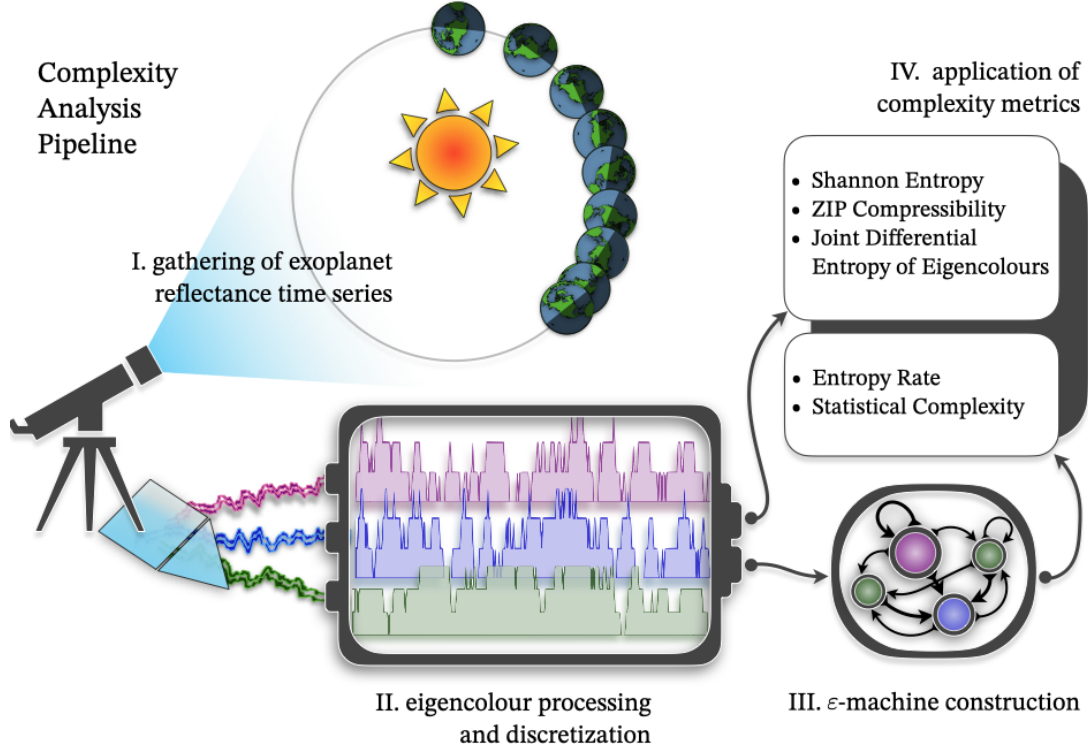


Figure 1. Analysis pipeline employed in the present work. Exoplanet reflectance time series, recorded for different wavelength bands, are the input data to our process (either from the DSCOVR mission for Earth, or the EMM for Mars). These time series are normalized and discretized before being analyzed by several methods: we use epsilon-machine reconstruction to compute the statistical complexity and entropy rate, we compute Shannon entropy using the standard formula, and use zip file compression to compute the zip compressibility ratio. The joint differential entropy is calculated according to the method presented in G. Segal et al. (2024).

2.2. Pre-processing of Time Series

The original DSCOVR Earth time series were resolved at the following wavelengths: 317, 325, 340, 388, 443, 551, 680, 688, 764, and 780 nm, and the Mars data were resolved at the following wavelengths: 260, 320, 437, 546, and 635 nm. Thus, we were only able to compare the following pairs with one another: 325 and 320 nm, 443 and 437 nm, and 551 and 546 nm. The data from Earth and Mars were also sampled at different rates (Earth higher than Mars) and taken over different durations (Earth longer than Mars). To control for these differences, we carried out the following processing steps:

1. Truncating the Earth series down to the same duration as Mars, such that both were 486.2 local days in length.
2. Compiling an evenly spaced time index from 0 to 486.2 at the average sampling rate of the Mars data (since this was lower than that for Earth).
3. Resampling all time series to the new time index to compensate for uneven sampling rates in the original data and to ensure that both sets of time series had equal final sampling rate. This was done using the MATLAB “interp1” function with linear interpolation. For the Mars data this process was a matter of small scale local interpolation (due to uneven sampling rate). For the Earth data it was a downsampling operation due to a higher original resolution.
4. Normalization of all time series to the unit interval [0,1].
5. Discretization to 40 levels within the unit interval. Note that higher or lower values did not significantly alter the

results (also see sensitivity analysis performed in our prior work (S. Bartlett et al. 2022)).

Example time series before and after pre-processing are shown in Figure 6.

2.3. Calculation of Information and Complexity Metrics

With the processed, discretized time series in hand, it was straightforward to compute the Shannon entropy of each using the standard formula:

$$H(X) = -\sum_{i=1}^n p(x_i) \log_2 p(x_i).$$

Computing the Zip compressibility ratio is also straightforward. We simply compressed the files containing the processed time series, and took the ratio of the compressed file size to the original file size. Technically this is the inverse of “compressibility,” but we used this ratio to retain the convention that higher complexity correlates with higher values. In this case higher inverse compressibility corresponds to higher algorithmic complexity (or the upper bound thereof).

Constructing epsilon machines (J. P. Crutchfield 2012) for the processed time series was performed using the “Decisional States” algorithm of N. Brodu (2011). We evaluated the statistical complexity and entropy rates (in units of bits per symbol) of these epsilon machines over a range of kernel width parameter values. This parameter controls the size of a coarse-graining kernel applied to finite sized windows (the size of this window is determined by the parameters “past

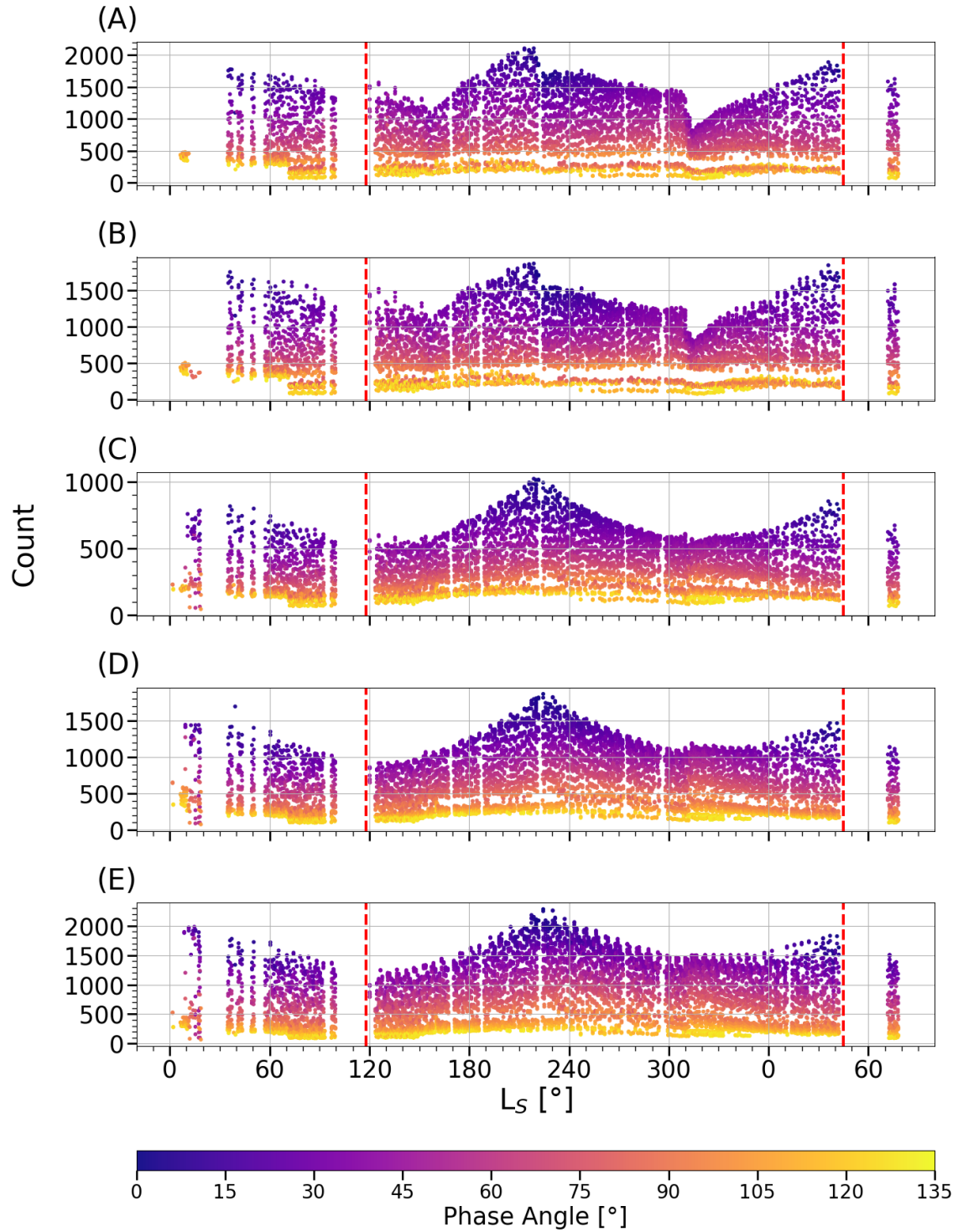


Figure 2. Original data related to the radiance of Mars, measured in Digital Number. The two vertical, dotted red lines indicate the range of the Martian time series selected for this study.

size” and “future size”) of the time series as the algorithm attempts to identify causal states. Very low values of kernel width lead to over-fitting to high frequency (often stochastic) features, and very high values lead to oversmoothing (epsilon machines with 0 entropy and complexity that have fit a single average value to the entire time series). Although one can readily choose a kernel width value that suits their specific purpose, in the present work we assessed both the maximum

value and the mean, taken over the range shown in Figure 7. This range was chosen because its lower limit coincides with the point where complexity is no longer changing with decreasing kernel width (the complexity and entropy rate curves remain constant with further decreases in kernel width), and the upper limit coincides with the point where complexity and entropy rate values have dropped to 0 (complete oversmoothing).

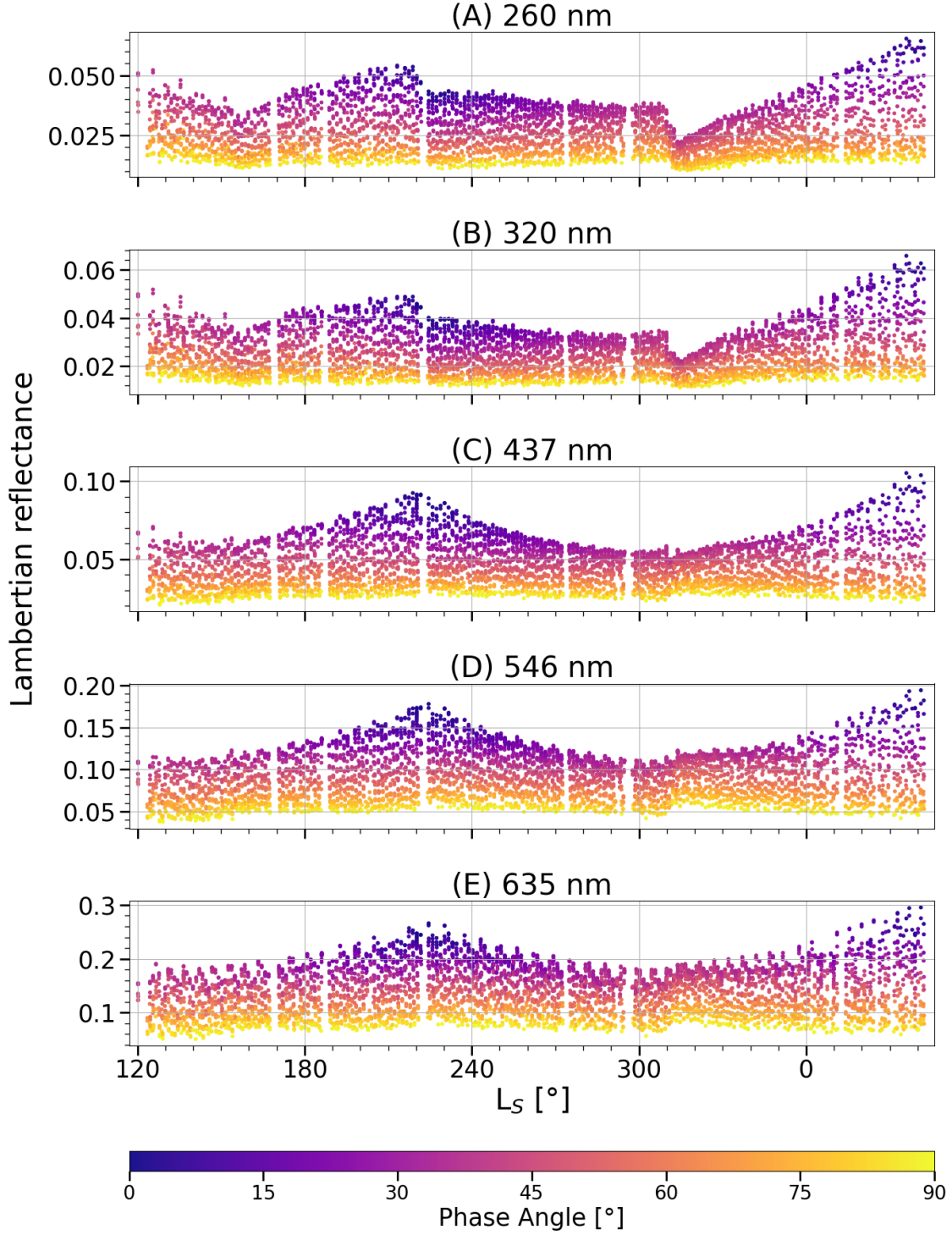


Figure 3. Reflectance converted from data in Figure 2, still significantly influenced by the phase angle.

To control for measurement uncertainties, we performed an error analysis upon both the Mars and Earth data. The Earth uncertainties were found to be negligible due to the fact that the time series are derived through disk integration of high resolution images. For the Mars data, errors were finite but relatively small. For each of the three wavelengths we generated 30 synthetic Mars time series, with each data point having a Gaussian noise term added according to the relative errors for each time point. The resulting uncertainties can be

seen in the error bars in Figure 7, the uncertainty values in Table 1 and the uncertainty ellipsoids in Figures 8 and 9.

We also computed JDE values for the set of 30 synthetic time series described above. The estimated JDE of eigencolours (i.e., the sum as represented by the plane in Figure 8) has a much larger absolute value than the standard error of the metric ($se = 0.123$) generated by producing a finite sampling distribution ($n = 30$) through the addition of Gaussian noise.

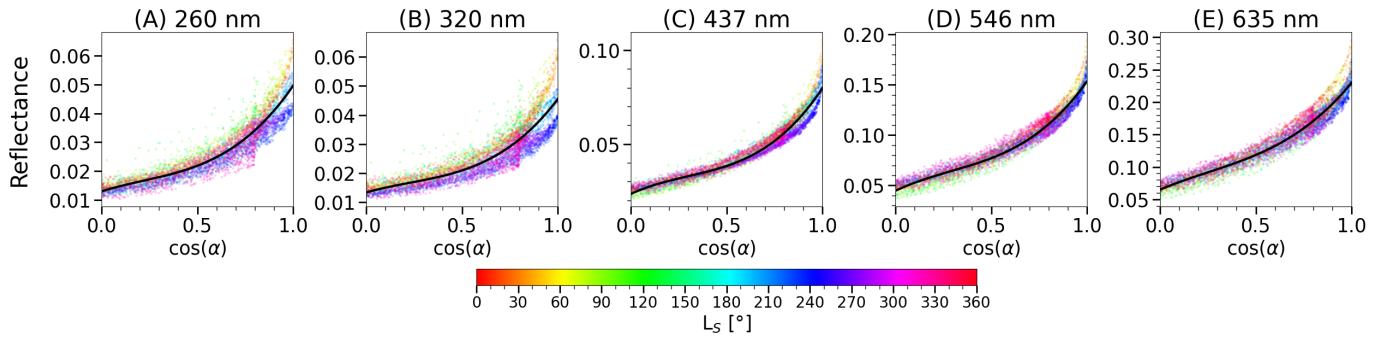


Figure 4. The fitting of Martian reflectance as a function of $\cos(\alpha)$, where α represents the phase angle. The color reflects the seasonal variation in reflectance.

Table 1
Informational and Complexity Metric Values for Earth and Mars Reflectance Time Series at Three Different Wavelengths

Metric	Mars 320 nm	Earth 325 nm	Mars 437 nm	Earth 443 nm	Mars 546 nm	Earth 551 nm
Shannon entropy	4.201 ± 0.083	5.018	4.535 ± 0.127	4.958	4.644 ± 0.118	4.920
Zip compressibility	0.1150 ± 0.0038	0.1548	0.1395 ± 0.0049	0.1641	0.1351 ± 0.0047	0.1678
Max stat' complexity	9.572 ± 0.185	11.162	10.168 ± 0.095	11.430	10.321 ± 0.113	11.425
Mean stat' complexity	4.039 ± 0.120	5.012	4.606 ± 0.178	4.983	4.715 ± 0.129	4.971
Mean entropy rate	1.129 ± 0.036	1.550	1.375 ± 0.058	1.635	1.332 ± 0.046	1.727

Note. Metrics shown are (a) Shannon entropy of the time series, (b) Zip compressibility ratio (compressed file size divided by original file size so that higher values correspond to higher algorithmic complexity), (c) Maximum statistical complexity as a function of kernel width, (d) Mean statistical complexity over kernel width, and (e) epsilon-machine entropy rate averaged over kernel width. Uncertainties are shown for Mars data but not for Earth since Earth uncertainties were negligible.

These results demonstrate the low overall error magnitudes and confirm that our conclusions are statistically significant.

3. Results

Our results are summarized in Figures 8 and 9. For all three wavelengths and all six metrics displayed, Earth exhibits higher values and its values cluster away from those of Mars.

Numerical values for all metrics applied are given in Tables 1 and 2.

The Shannon entropy is an elementary statistical measure that, crudely speaking, evaluates the spread within a probability distribution (in this case of the discrete time series values). Note that this measure is not sensitive to the temporal sequence of the input data, it simply assesses the spread in a given probability distribution.

Earth having higher Shannon entropy values implies that its normalized reflectance values are more broadly spread over the unit interval than those of Mars. One cannot definitively ascribe this to higher stochasticity, since deterministic time series can also exhibit high Shannon entropy. However, in this case, given the mix of both deterministic (e.g., the diurnal cycle) and stochastic (e.g., turbulent atmospheric fluctuations) features, we can speculate that Earth's time series is likely "more noisy" and less predictable than those of Mars.

The zip compressibility is a powerful upper bound for the algorithmic or Kolmogorov complexity. Given a string of symbols, the Kolmogorov complexity is the length of the shortest program, which, when executed by a Turing machine, produces the string. Hence, it represents the information required to encode the generating mechanism of a given process or set of measurements, assuming the Turing machine as the mode of computation. Since computing the Kolmogorov

complexity is technically undecidable (due to the Halting problem; A. Church 1936), it can only be estimated from above (H. Zenil et al. 2023). The zip compressibility thus offers a convenient means to perform this task, since compressing data files is a primary goal of computer science and hence a mature field. Many modern algorithms for compression, including zip, make use of the Lempel–Ziv 77 algorithm (J. Ziv & A. Lempel 1977) and the Huffman encoding (D. A. Huffman 1952). These algorithms allow the zip process to find repeating patterns and redundancies in data (even in a hierarchical fashion), then find more compact ways to encode and represent such features.

In Figure 8, the inverse zip compressibility ratio (compressed file size as a fraction of original file size) is plotted on the y-axis, such that an incompressible (maximum algorithmic complexity) file would have a value of 1, and a highly compressible file would tend to a value of 0. Hence, Earth's reflectance time series were less compressible than those of Mars.

The JDE for planetary reflectance time series, developed in G. Segal et al. (2024), aims to remove redundancies between channels or signals, by first performing a principal component analysis, producing the so-called "eigencolours" of the data. These eigencolours are linear recombinations of the original time series (multi-channel, in our case multiwavelength). The eigencolours are ranked in order of the degree to which they represent variability in the data. The advantage of this approach is that variability that is present across multiple channels does not get counted more than once when averaging across channels. From the time series of eigencolours, we compute estimates of the differential entropy, the entropy corresponding to a continuously valued distribution (as opposed to discrete, binned data, as is conventional). Since

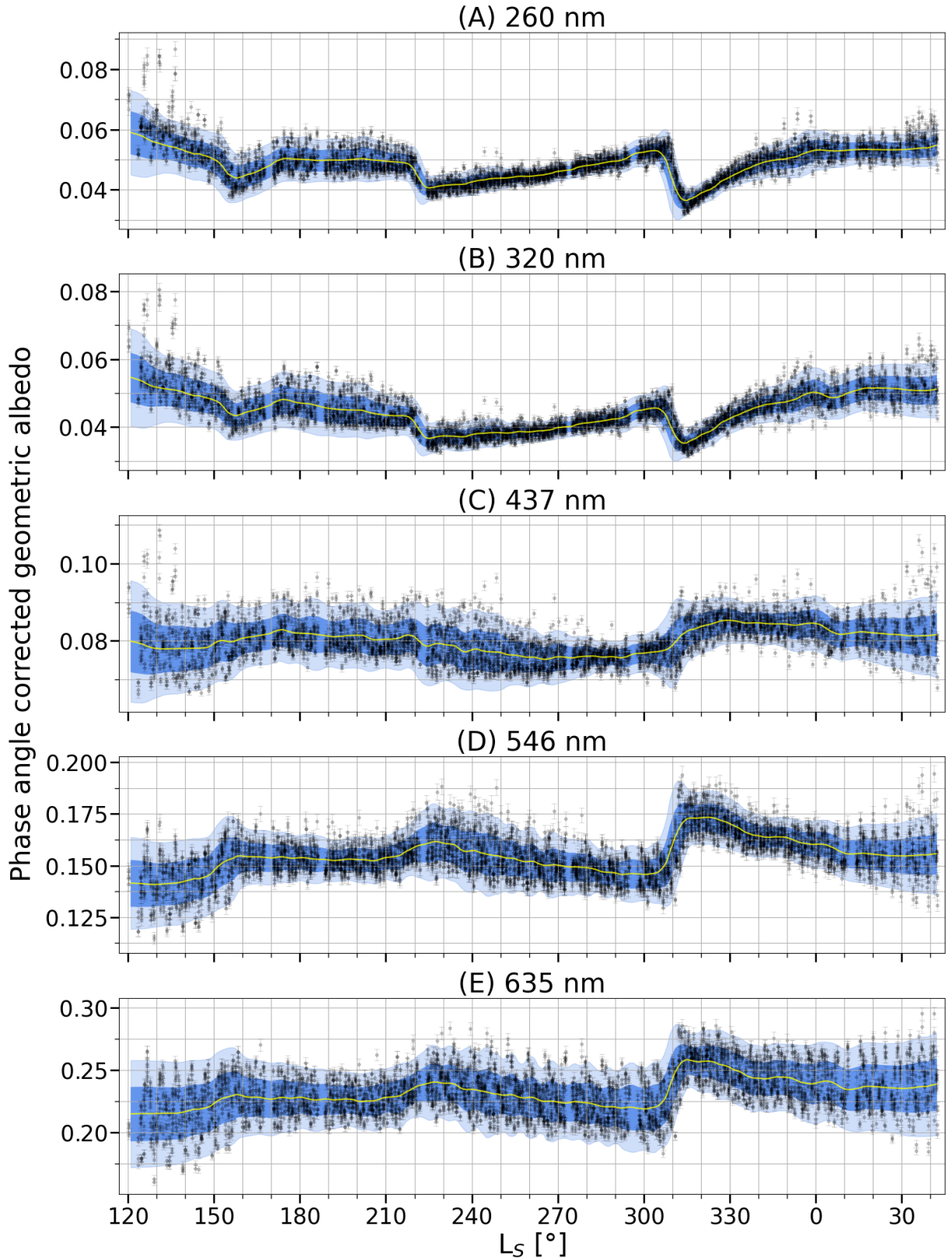
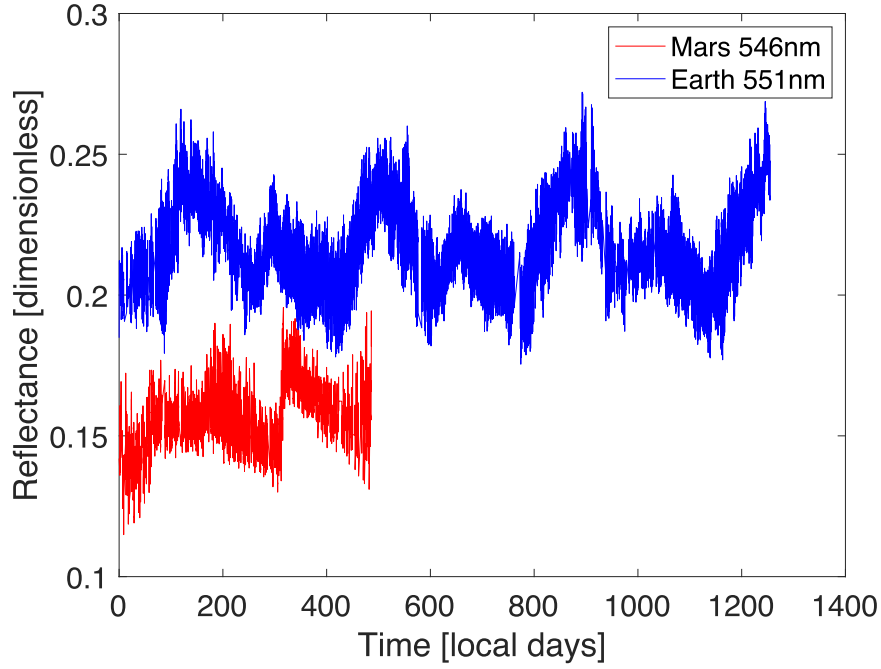


Figure 5. Reflectance corrected for phase angle, now influenced only by variations in surface composition.

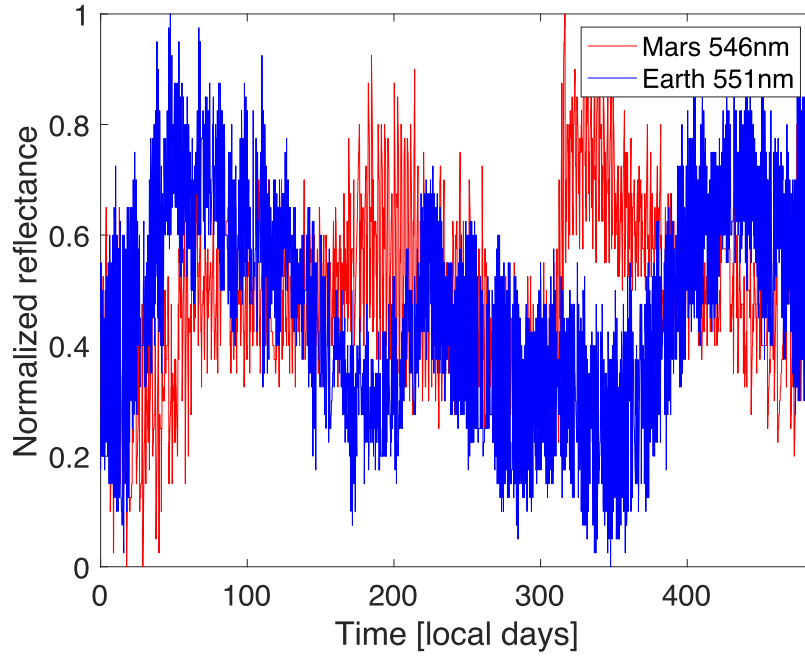
the strength of the JDE lies in its ability to capture all the variability in a set of time series without overcounting repeated features, we present the average JDE of the three eigencolours derived from the three wavelength channels analyzed (plotted

as horizontal planes in Figure 8). As with the other metrics, Earth ranks higher than Mars.

On the vertical axis of Figure 9 the maximum statistical complexity is plotted (maximum value as a function of kernel



(a) Original time series before processing.



(b) Time series after normalisation and discretisation.

Figure 6. Example time series used in our analysis. The top panel shows the original data for Earth at 551 nm and Mars at 546 nm. Note the different overall lengths (sampling rates are also different, with Earth having a slightly higher rate). The lower panel shows the same time series after normalization, truncation of the Mars series, and resampling to the same, equally spaced time index.

width, see Section 2.3). This metric bears a similarity to algorithmic complexity, in the sense that it aims to measure the size of the smallest generating mechanism or model of a given process. The primary difference between statistical complexity and Kolmogorov complexity is that epsilon machines (from which statistical complexity is calculated) are in general stochastic, whereas Turing machine programs and Turing machines are completely deterministic. Thus, the algorithm of epsilon-machine reconstruction is designed to distinguish

noise and data and represent them appropriately in the resulting generative model (J. P. Crutchfield 2012).

Our results imply that the minimal, predictive epsilon-machine models of the time series had a higher state space information content for Earth than Mars. Epsilon-machine entropy rates and statistical complexities taken as a mean over kernel width values (as opposed to maxima) were also higher for Earth than Mars (these are shown on the horizontal axes of Figure 9).

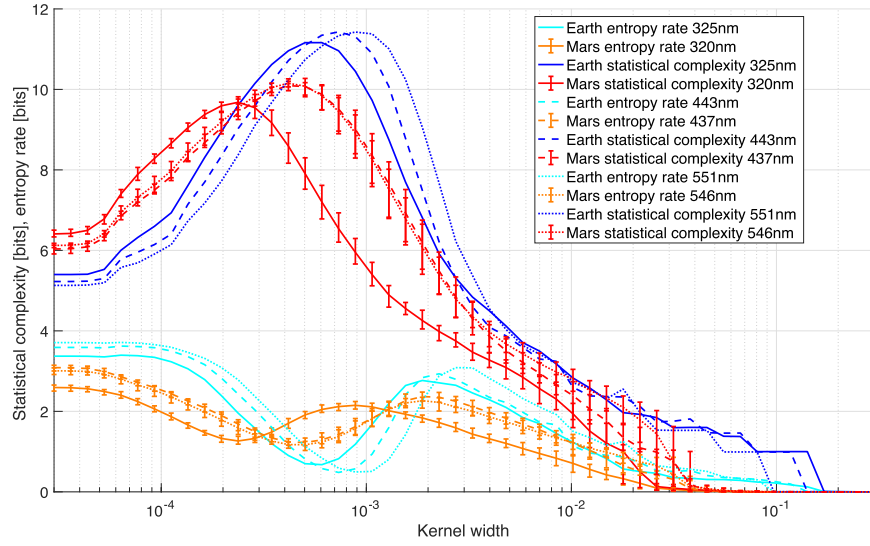


Figure 7. Statistical complexity and entropy rate curves over kernel width parameter values. Earth complexities are shown in blue and entropy rates in cyan. Mars complexities are shown in red and entropy rates in orange. Solid lines correspond to wavelengths ~ 322 nm, dashed lines to ~ 440 nm, and dotted lines to wavelengths ~ 550 nm. Error bars are shown for Mars metrics but not Earth since Earth uncertainties were found to be negligible.

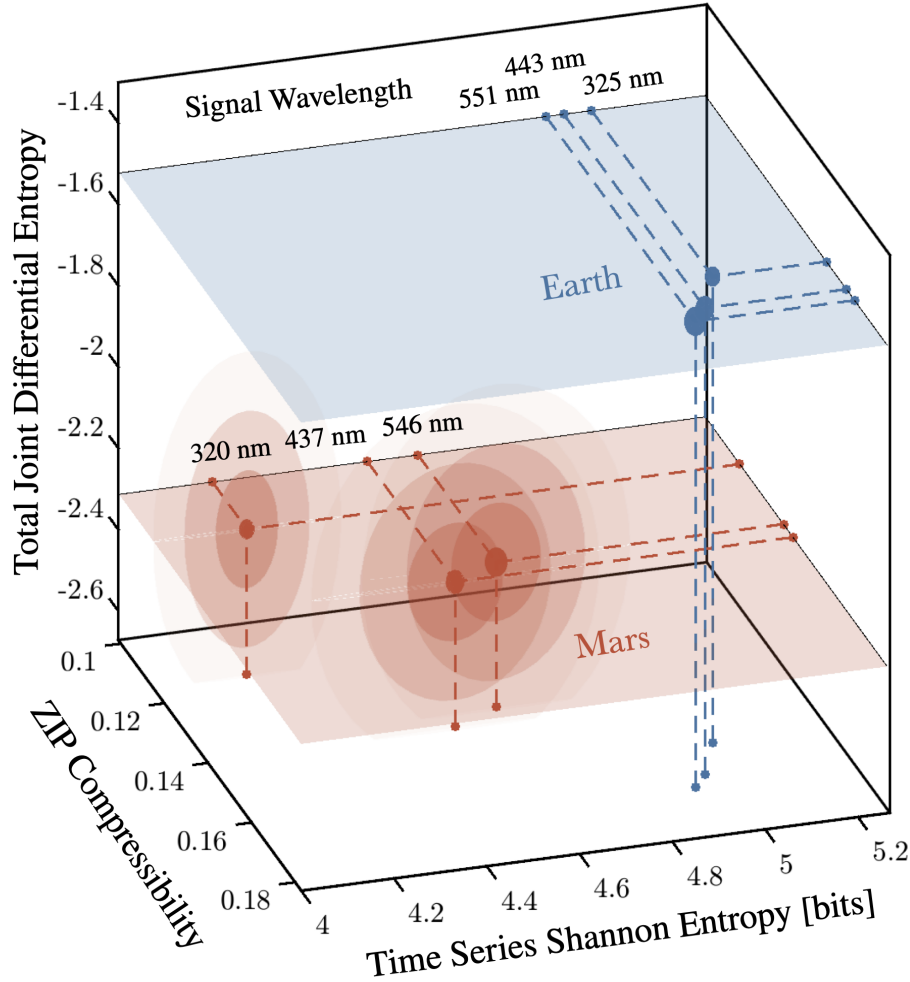


Figure 8. Entropy and compressibility results for Earth and Mars reflectance time series at three different wavelengths. Metrics shown are the Shannon entropy of the time series on the x-axis, ZIP compressibility ratio (compressed file size divided by original file size so that higher values correspond to higher algorithmic complexity) on the y-axis and JDE on the z-axis. Horizontal and vertical lines are plotted emanating from each point to aid visualization, in addition to translucent planes showing the heights of the summed JDE values for Earth and Mars, respectively. Ellipsoids emanating from Mars data points correspond to the respective uncertainties (Earth uncertainties were found to be negligible; see Section 2.3 for details).

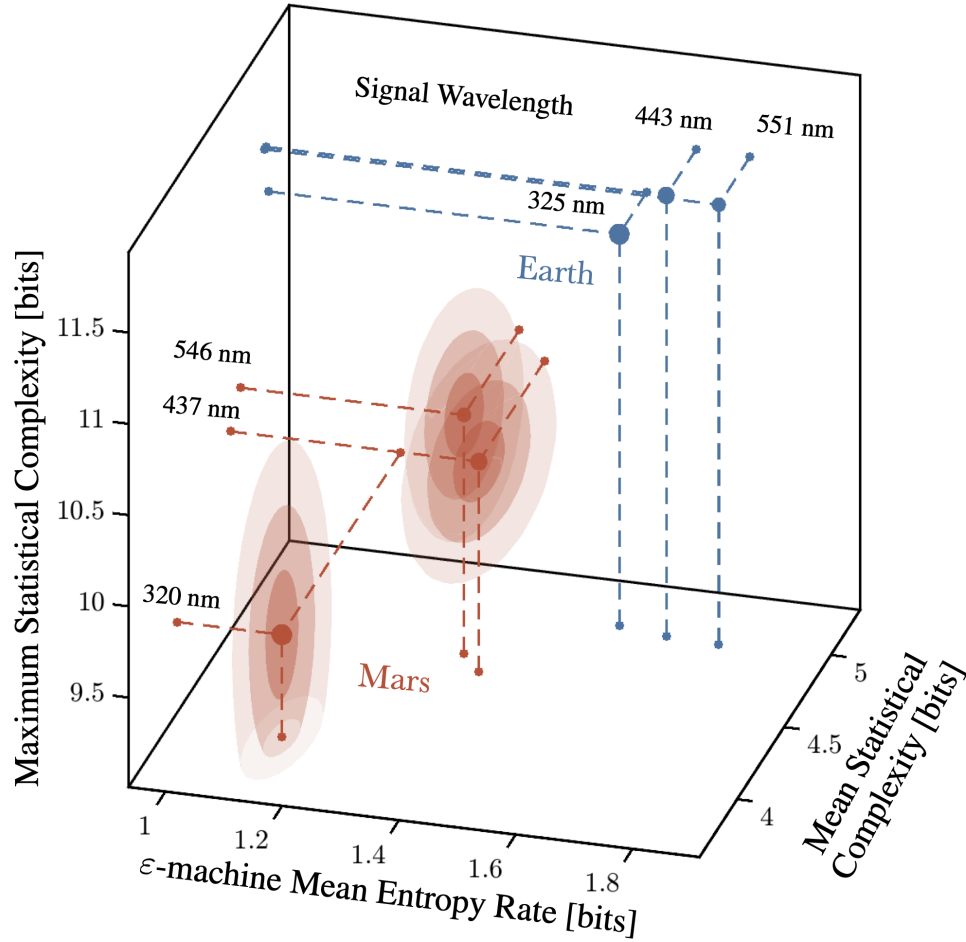


Figure 9. Results from computational mechanics analysis of Earth and Mars reflectance time series at three different wavelengths. Metrics shown are (a) mean statistical complexity on the x -axis, (b) mean epsilon-machine entropy rate on the y -axis, and (c) maximum statistical complexity (as a function of kernel width) on the z -axis. Horizontal and vertical lines are plotted emanating from each point to aid visualization. Ellipsoids emanating from Mars data points correspond to the respective uncertainties (Earth uncertainties were found to be negligible, see Section 2.3 for details).

Table 2
JDE Results for Renormalized, Discretized Reflectance Time Series for Earth and Mars

Metric	PC1		PC2		PC3		Sum	
	Mars	Earth	Mars	Earth	Mars	Earth	Mars	Earth
Eigenvalue	0.040	0.110	0.020	0.008	0.001	0.001	0.060	0.119
Diff' entropy	-0.53	-0.38	-1.02	-0.57	-0.75	-0.56	-2.31	-1.52
Variance	0.0161	0.0456	0.0206	0.0398	0.0238	0.0331	0.060	0.119

Note. Metrics shown are eigenvalues, differential entropy, and variance for each principal component and their sums.

4. Conclusions

Exoplanets fundamentally challenge our scientific and philosophical ideas. They are alluring marvels, adrift across vast reaches of space. Yet, the faint trickle of photons they send our way offer the chance to unlock intricate details about their nature. Though much of exoplanetary science rests on single, static glimpses, our work reveals that features of these distant worlds can be revealed when observed through the lens of time. These dynamic signals not only enrich our understanding but may even inform our credence as to whether planets might harbor life—or have the potential to nurture it (S. Bartlett et al. 2022; M. L. Wong et al. 2022).

In the present work we assess whether such streams of photons from Earth, and our sister planet, Mars, could be used to distinguish the fact that Earth is inhabited while Mars sits lifeless. In pursuit of this goal, we applied a suite of informational and complexity metrics to planetary reflectance time series for Earth and Mars, derived from disk-integrated images of the planets, resolved over time. These metrics include the Shannon entropy, Zip compressibility, JDE, statistical complexity, and epsilon-machine entropy rate. We found that Earth exhibited higher values for all metrics employed, supporting the hypothesized association between life and planetary complexity (S. Bartlett et al. 2022). We note that these metrics are overlapping by definition, in terms of the

statistical and informational features they detect in time series data. Each has strengths, weaknesses and favored applications, and we display them all to illustrate the consistent finding that complexity appears to correlate with the presence of life (S. Bartlett et al. 2022). That being said, future work will perform dimensionality reduction analyses to quantify the degree of redundancy between these measures (the overlap of statistical features that the measures are sensitive to). This will allow us to condense the method to 1 or 2 key metrics or metric combinations.

Given the similar results yielded by the different metrics, it is natural to ask why it is necessary to employ the more sophisticated metrics such as statistical complexity or JDE, if simpler measures like the Shannon entropy can make the same distinctions. The reason we make use of the more advanced metrics is in anticipation of scenarios where simpler metrics are unable to distinguish important features due to their intrinsic limitations and potential for false positives or false negatives. For example, the temporal sequence of measurement values is important for epsilon machines, but irrelevant to the Shannon entropy. Thus, we foresee future applications (e.g., technosignatures) where different signals may exhibit the same Shannon entropy values, but different statistical complexities, for example. Indeed, one could produce ensembles of time series where different members of the set have different statistical complexities, but equal Shannon entropies.

It is likely that Earth’s high-complexity stems from its intricate hydrological cycle and atmospheric processes, as well as its diverse surface contrasts (e.g., between land and ocean). As with many biosignature approaches, our method scrutinizes abiotic features that could serve as proxies for the presence of life, and in such a situation it is difficult to affirm a causal link between life and the observed features. Rather, there are two potential causal explanations for the observations: (1) Earth’s complexity is a direct result of the biosphere and its myriad, bidirectional interactions with the planet; or (2) Earth is inherently complex, and would be equivalently so even if life had never originated. Until we find a bona fide sterile, “Earth twin” (or have a high fidelity simulation of such a world), this question will remain open. Further discussions of these fascinating questions can be found in S. Bartlett et al. (2022), H. B. Smith & C. Mathis (2023), M. L. Wong et al. (2022), and references therein.

In the meantime, this work continues to test the hypothesis that only biology can cause a planet to become as complex as Earth. We contend that it is highly unlikely that Earth’s biosphere is simply “along for the ride” and has little influence or feedback with its environment. Evolutionary biology and ecology have long established the diverse effects that life has on its surroundings, by various mechanisms such as niche construction (F. Odling-Smee et al. 2013) and influencing geochemical cycles (P. G. Falkowski et al. 2008; T. Lenton & A. Watson 2013), among myriad others. Future work will explore the use of general circulation models (GCMs) to perform interventional experiments on Earth’s atmospheric and biogeochemical features. Specifically, the measured complexity will be tracked during the removal or reduction of biogeochemical influences. This will permit the estimation of the influence of biotic effects upon measured planetary complexity. We are also in the process of expanding our analysis to additional solar system bodies including Venus and

Titan, as well as synthetic exoplanets simulated using exoplanet GCMs.

We do not suggest that our presented method is the final answer to life detection, but rather propose it as a distinct but powerful complement to other methods rapidly developing in the field. All such methods must carefully consider possible false positives. In the case of the method presented herein, potential false positives could arise from complex but sterile worlds, where abiotic features such as turbulence, nonlinear growth processes (e.g., complex crystal structures), or nonlinear pattern-forming chemical systems give rise to high-complexity reflectance signatures. This reinforces the need for multiple, complementary approaches in the search for life. Humanity’s first detection of any such life will have to lean upon many or all of our arsenal of techniques and observations.

In addition to complexity being expressed by nonliving phenomena, complexity measures cannot explicitly identify meaning, function or semantics in a system (whether living or nonliving). This was noted by Shannon himself when formulating his famous theory of communication (C. E. Shannon 1948). Recent work has however, begun to try and fill this gap, within the theory of “Semantic Information Theory” (SIT; A. Kolchinsky & D. H. Wolpert 2018; D. R. Sowers et al. 2023). The premise of SIT is that a given quantity of Shannon information (e.g., a bit string or sample of DNA) may carry differing amounts of meaning, significance or semantics. In the case of “viability-based” approaches to SIT (A. Kolchinsky & D. H. Wolpert 2018), meaning and semantics of a given quantity of information are directly correlated with the influence that that information has on an agent’s viability (its ability to survive and persist in a given environment). In more recent approaches, meaning and semantics are associated with the degree to which information can influence a system’s future trajectories through its phase space (the set of degrees of freedom required to identify the physical state of that system; D. R. Sowers et al. 2025). This growing field of SIT and its methods are now being leveraged in the search for life (S. Bartlett et al. 2025; D. R. Sowers et al. 2025). The overall hypothesis being pursued is that life will tend to exhibit higher levels or concentrations of semantic information than abiotic systems. As with complexity metrics, SIT could offer approaches that are complementary to existing biosignature methods, such that humanity’s overall probability of finding life elsewhere continues to be enhanced.

Time-resolved reflected-light photometry is feasible now for close-in giants with JWST and will be routine for temperate terrestrials with the HWO. JWST’s time-series modes (NIRCam/MIRI) have demonstrated sub-0.1% precision and stable, orbit-long phase curves in which the near-IR reflected component can be isolated for highly irradiated planets, yielding bona-fide reflectance time series. HWO, by contrast, is expressly designed to directly image ≥ 25 Earth-size habitable-zone planets at $\sim 10^{-10}$ contrast, enabling repeated broadband observations that produce rotational and seasonal reflectance light curves. Established inversion techniques (spin-orbit tomography; Earth EPOXI/DSCOVR benchmarks) could then recover physically interpretable maps—ideal inputs for our reflectance-complexity framework.

Hence we suggest that new phyla of exoplanet and life detection missions also examine time series, lest the intrinsic dynamic properties of exoplanets might go uncharted, despite hiding in plain sight.

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Author Contributions

S.J.B. conducted the information and complexity analysis, and wrote the text. X.S. and S.F. processed and curated the Mars time series data. G.S. conducted the JDE analysis, assisted with writing the text, and making the figures. D.R.S. produced the main figures, and provided proof-reading and editing support. Y.L.Y. provided guidance and support.

Author Declarations

Authors declare no competing interests.

ORCID iDs

Stuart Bartlett  <https://orcid.org/0000-0001-5680-476X>
 Xianlei San  <https://orcid.org/0009-0004-6304-2600>
 Siteng Fan  <https://orcid.org/0000-0002-3041-4680>
 Gary Segal  <https://orcid.org/0000-0001-7165-8860>
 Damian R. Sowinski  <https://orcid.org/0000-0003-3136-7449>
 Yuk L. Yung  <https://orcid.org/0000-0002-4263-2562>

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